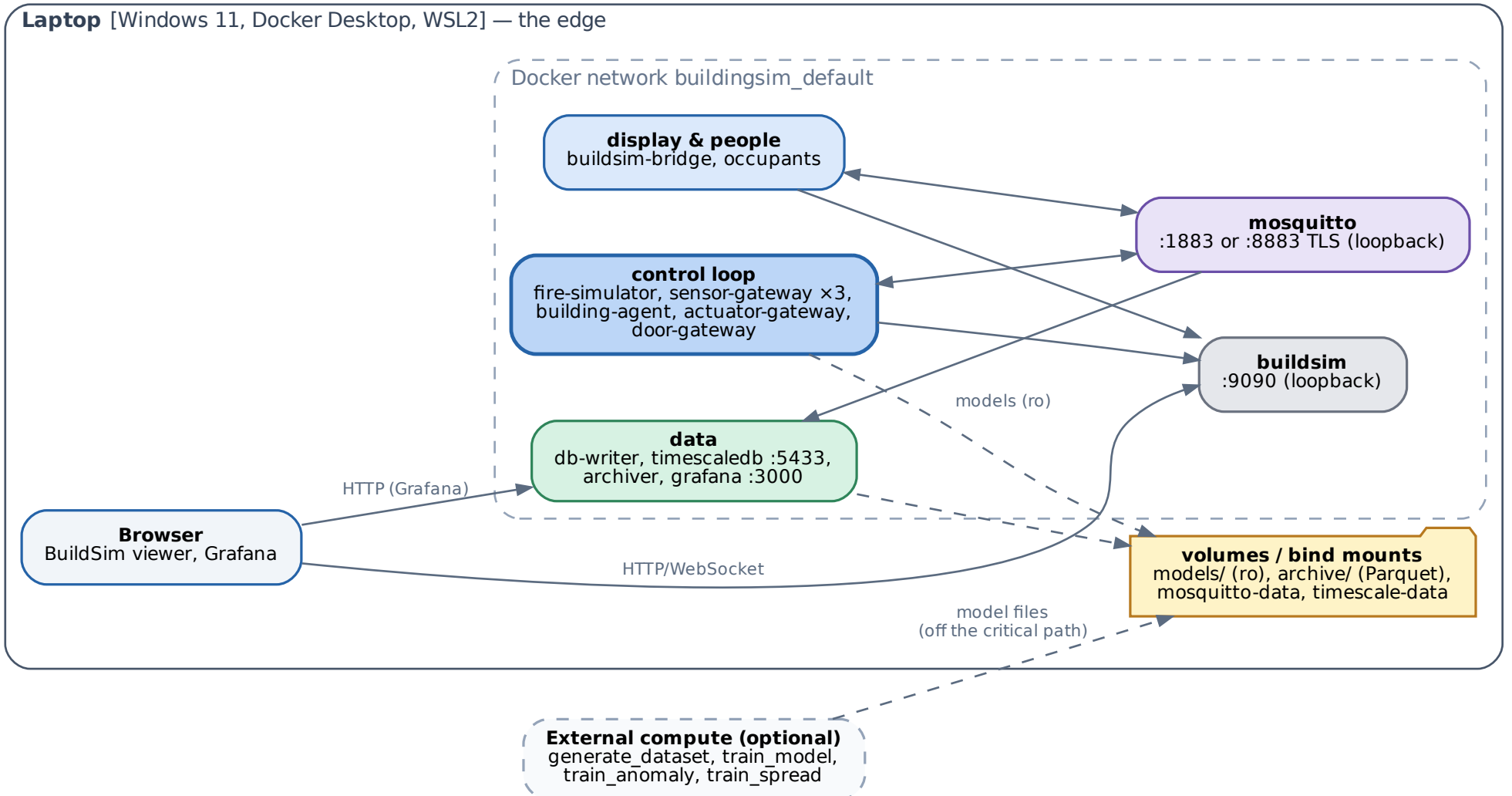
Laptop [Windows 11, Docker Desktop, WSL2] — the edge
Docker network buildingsim_default
display & people
buildsim-bridge, occupants
mosquitto
:1883 or :8883 TLS (loopback)
control loop
fire-simulator, sensor-gateway ×3,
building-agent, actuator-gateway,
door-gateway
buildsim
:9090 (loopback)
models (ro)
data
db-writer, timescaledb :5433,
archiver, grafana :3000
HTTP (Grafana)
Browser
BuildSim viewer, Grafana
HTTP/WebSocket
volumes / bind mounts
models/ (ro), archive/ (Parquet),
mosquitto-data, timescale-data
model files
(off the critical path)
External compute (optional)
generate_dataset, train_model,
train_anomaly, train_spread